SANJAY DIGITAL WORKS

Progression

Complete Project Reference

Every file: what it does, why it exists, and when to change it.
Rules, data, build steps, known issues and how-to recipes.

REPOSITORY	github.com/sanjay200k/progression
BRANCH	main
COMMIT	ac7343d
APP ID	com.sanjaydigitalworks.progression
WRITTEN	11 October 2026
STACK	Vanilla JS PWA + Capacitor 7 Android + Supabase (optional)

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1. How to use this document

Open this PDF when you want to change something in Progression. Go to section 3 (Change guide) first: it tells you which file to edit. Then read that file's entry in section 4 to see what it does and what can break.

Reading order

- Section 2: the big picture and folder map.
- Section 3: I want to change X, so edit file Y.
- Section 4: every file, one entry each.
- Section 5: data and game rules (what the numbers mean).
- Section 6: step-by-step build, test, release procedures (Windows cmd).
- Section 7: known issues and technical debt.
- Section 8: roadmap status and glossary.

Every file entry has the same rows

Row	Meaning
What it does	Plain description of the file.
Why it exists	The reason the file is in the project.
What is inside	Main functions, settings and values.
Edit when	The situations where you should change it.
Caution	Things that break or are risky (shaded amber).

2. Project overview

Progression is a gamified goal and habit tracker. The user creates goals, breaks them into daily missions, and earns XP, levels, ranks, streaks and achievements. It also has trackers (sleep, water, any custom number) with trend charts.

It is a vanilla JavaScript web app (no framework, no bundler, no build step for the web code). The same web code runs three ways: in a browser as a PWA, inside an Android WebView app built with Capacitor, and in unit tests under Node (only the pure rule modules).

All data lives on the device in localStorage. Optional cloud sync uses the user's own Supabase project over plain REST calls. There is no custom server.

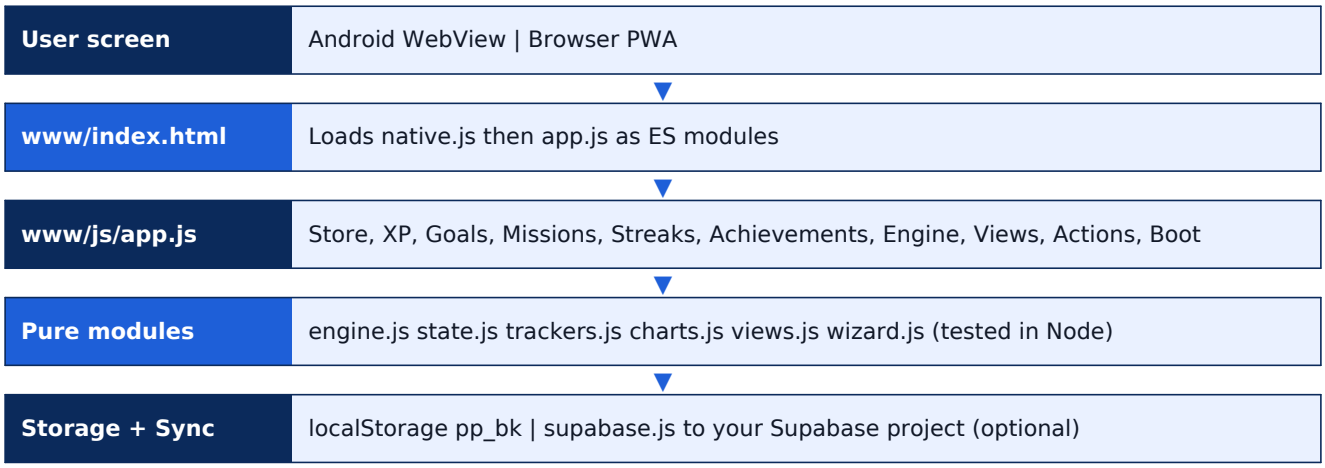
Identity

App name	Progression
Android app ID	com.sanjaydigitalworks.progression (cannot change after Play Store publish)
Repository	github.com/sanjay200k/progression (branch main)
Local folder	F:\!DigitalProduct\DailyActivityWebsite_mobile app\progression
Version in app	versionCode 1, versionName 1.0 (android/app/build.gradle)
Document written	11 Oct 2026, repo at commit ac7343d

Technology

Part	Detail
Language	JavaScript (ES modules, no TypeScript), HTML, CSS. Java only for the empty Android MainActivity.
UI	Hand-written template strings rendered into #app with innerHTML. Event delegation via data-a attributes.
Native shell	Capacitor 7 (Android). Web files in www/ are copied into the Android project by npx cap sync.
Storage	Browser localStorage (key pp_bk). Daily snapshots in pp_snapshots.
Cloud (optional)	Supabase REST + email OTP login. Table progression_state with row level security.
Charts	Pure SVG generated by charts.js and app.js. No chart library.
Fonts	Chakra Petch and IBM Plex Sans, self-hosted woff2 in www/fonts/.
Tests	Node built-in test runner. tests/merge.test.mjs and tests/engine.test.mjs (15 tests pass).
CI	GitHub Actions: test, debug APK, optional signed release AAB.
Android build	Gradle 8.11.1, Android Gradle Plugin 8.7.2, compileSdk and targetSdk 35, minSdk 23, Java 21.

How the pieces connect



From edit to installed app

1. Edit web code in www/ (HTML, CSS, JS).
2. Run tests: npm test (needs Node.js installed).
3. Commit and git push to main.
4. GitHub Actions runs: test, then debug APK build (and signed AAB if secrets exist).
5. Download the APK artifact from the Actions run, install on phone.
6. Before each Play Store upload: raise versionCode in android/app/build.gradle.

Folder map

```
progression/
|-- .github/workflows/android.yml     CI: test, debug APK, signed AAB
|-- android/                          Generated Capacitor Android project
|   |-- app/
|   |   |-- build.gradle              App build: IDs, version, R8, signing
|   |   |-- capacitor.build.gradle    GENERATED by cap sync - do not edit
|   |   |-- proguard-rules.pro        R8 keep rules (empty template)
|   |   `-- src/main/
|   |       |-- AndroidManifest.xml   Permissions, activity, provider
|   |       |-- java/.../MainActivity.java
|   |       `-- res/                  Icons, splash, strings, styles
|   |-- build.gradle                  Top-level Gradle (AGP version)
|   |-- variables.gradle              SDK and library versions
|   |-- settings.gradle / capacitor.settings.gradle
|   |-- gradle.properties             JVM memory, AndroidX flag
|   |-- gradlew, gradlew.bat, gradle/wrapper/   Gradle launcher
|   `-- .gitignore
|-- tests/
|   |-- merge.test.mjs                Cloud merge + backup validation
|   `-- engine.test.mjs               Streak, freeze, rest day, rank rules
|-- www/                              THE WEB APP (what users see)
|   |-- index.html                    Shell + security policy
|   |-- styles.css                    All styling and themes
|   |-- manifest.webmanifest          PWA install metadata
|   |-- sw.js                         Service worker (offline cache)
|   |-- icon-192.png, icon-512.png, icon-maskable-512.png, apple-touch-icon.png
|   |-- fonts/                        6 woff2 font files
|   `-- js/
|       |-- app.js                    Main app (84 KB, one big file)
|       |-- engine.js                 Pure rule functions
|       |-- state.js                  Validation, merge, snapshots
|       |-- trackers.js               Tracker logic
|       |-- charts.js                 SVG trend chart
|       |-- supabase.js               Cloud sync client
|       |-- views.js                  safeText, percent helpers
|       |-- wizard.js                 Goal length constants
|       `-- native.js                 Capacitor hooks (status bar, back)
|-- capacitor.config.json             App ID, name, web folder
|-- package.json / package-lock.json  npm scripts and dependencies
|-- README.md  CHANGELOG.md  .gitignore
```

3. Change guide: I want to... so edit...

Find your task in the first column. File and detail columns tell you where and what to look for.

I want to change	Edit this file	Where / how
Change XP per mission (Easy/Standard/Hard)	www/js/app.js	CFG.diffXP = {1:40, 2:75, 3:120}
Change daily, milestone, goal bonus XP defaults	www/js/app.js	CFG.bonus = {day:100, ms:500, goal:2000}. Users can override in Settings.
Change level curve	www/js/app.js	CFG.need(n) = round(1000 * 1.18^(n-1) / 50) * 50
Change rank letters (E to S) and level cutoffs	www/js/engine.js	rankForLevel()
Add or edit an achievement	www/js/app.js	ACH array. Row: [id, name, description, icon, xp, metric, minimum, category]. New category: also add it to ACHIEVEMENT_CATEGORIES in engine.js.
Add a new achievement metric	www/js/app.js	Ach.metrics() returns the metric values. Add your key there, then use it in ACH.
Change streak freeze rule (1 per 14 best-streak days)	www/js/engine.js	freezeAllowance() and freezeDatesForHistory()
Change rest day limit (no other rest day within 6 days)	www/js/engine.js	canAddRestDay() and the message text in app.js action 'rest'
Change what counts as a perfect day	www/js/engine.js	isPerfect()
Change default goal length (90) or max (730)	www/js/wizard.js	DEFAULT_GOAL_DAYS, MAX_GOAL_DAYS. Error text in app.js Wiz.next().
Change colors, spacing, fonts, themes	www/styles.css	:root block (tokens --bg, --ac, --ok, --xp ...). Light theme blocks follow it.
Change app name shown on phone	android/app/src/main/res/values/strings.xml + capacitor.config.json	app_name, title_activity_main, appName
Change app icon / splash	android/app/src/main/res/ (mipmap-*, drawable*) and www/icon-*.png	Replace PNGs keeping same file names and sizes.
Change PWA install name/colors	www/manifest.webmanifest	name, short_name, theme_color, background_color
Change tracker colors or types	www/js/trackers.js	TRACKER_COLORS, validTypes; keep the same lists in state.js validateBackup().
Change chart look	www/js/charts.js (trackers) / app.js View.chart (XP and score)	SVG built as template strings. Styles in styles.css .tracker-* and .chart*.
Add a new screen/page	www/js/app.js	Add entry to NAV array, add a function View.myPage(), links use data-a=go data-r=myPage.
Add a new button action	www/js/app.js	Add data-a=name in markup, add else-if branch in the document click handler near the end of file.
Change Supabase table or fields	www/js/supabase.js + README.md SQL	REST URL /rest/v1/progression_state. Change SQL in Supabase dashboard too.
Add a new JS file	www/sw.js + www/js	Add path to APP_SHELL in sw.js and bump RELEASE_CACHE so offline browsers update.

I want to change	Edit this file	Where / how
Release a new web version for browser users	www/sw.js	Change RELEASE_CACHE from progression-v6 to v7 etc.
Release a new Android version	android/app/build.gradle	Raise versionCode by 1 (must always increase) and update versionName.
Add a native Capacitor plugin	package.json, then terminal	npm install @capacitor/xxx, then npx cap sync android, commit package-lock.json.
Change Android permissions	android/app/src/main/AndroidManifest.xml	uses-permission lines at the bottom.
Change min/target Android version	android/variables.gradle	minSdkVersion, targetSdkVersion, compileSdkVersion
Change CI behaviour	.github/workflows/android.yml	Jobs: test, debug-apk, release-aab.
Add or change tests	tests/*.test.mjs	Use node:test and node:assert/strict. Run with npm test.

4. File-by-file reference

4.1 Root files

`package.json`	842 B
What it does	npm project file. Lists scripts and library versions.
Why it exists	Lets anyone (and GitHub Actions) install the exact tools with one command.
What is inside	• Scripts: serve (python web server on port 4173), test (node --test, finds tests/*.test.mjs), android:add, sync (cap sync), open (cap open android), apk (sync + gradlew assembleDebug), aab (sync + bundleRelease), apk:release (sync + assembleRelease). • Dependencies (all ^7): @capacitor/core, @capacitor/android, @capacitor/status-bar (used), @capacitor/app (used by native.js back button), @capacitor/haptics, @capacitor/local-notifications, @capacitor/preferences (installed but NOT used by app code yet). • devDependency: @capacitor/cli. "private": true so it is never published to npm.
Edit when	Edit when adding a plugin, a script, or bumping the app version string.
Caution	After any dependency change run npm install and commit package-lock.json, or CI fails at npm ci. Missing "type":"module" causes a harmless Node warning when tests run.

`package-lock.json`	42 KB
What it does	Exact version of every installed library and its sub-libraries.
Why it exists	Makes builds repeatable. CI uses npm ci which refuses to run if this file does not match package.json.
What is inside	• Generated by npm. Never edit by hand.
Edit when	Regenerate with npm install (or npm install --package-lock-only) whenever package.json dependencies change.
Caution	Commit it every time it changes.

`capacitor.config.json`	188 B
What it does	Capacitor settings: app ID, app name, web folder, WebView scheme, background color.
Why it exists	Tells Capacitor which folder is the web app (www) and what the Android app is called.
What is inside	• appId = com.sanjaydigitalworks.progression • appName = Progression • webDir = www • server.androidScheme = https (app runs at https://localhost inside the WebView) • android.backgroundColor = #0a0f1d (dark navy, shown before the page paints)
Edit when	Edit appName or backgroundColor freely. After editing run npx cap sync android.
Caution	Do NOT change appId after Play Store publish. Do NOT change androidScheme later: it changes the page origin and users lose their localStorage data.

`.gitignore`	97 B
What it does	Tells git which files to never commit.
Why it exists	Keeps junk and machine-specific files out of the repo.
What is inside	• node_modules/, android/.gradle/, android/build/, android/app/build/, android/local.properties, .idea/

Edit when	Add more patterns when new secret or generated files appear.
Caution	SECURITY GAP: *.jks, *.keystore and keystore.properties are NOT ignored (the lines in android/.gitignore are commented out). Add them before creating a release keystore.

`README.md` 2.8 KB

What it does	Short developer guide.
Why it exists	First file a new developer reads.
What is inside	• How to run locally (python -m http.server 4173 --directory www). • Data keys (pp_bk, pp_snapshots) and a manual test checklist. • Supabase SQL: progression_state table with 3 row-level-security policies. • Sync merge notes and how to build the APK in the cloud.
Edit when	Keep in sync with the real commands.
Caution	STALE: says node tests/logic.test.mjs but that file no longer exists. Real command is npm test.

`CHANGELOG.md` 6.6 KB

What it does	History of 41 numbered fixes and features from earlier passes.
Why it exists	Explains why code looks the way it does (comeback rule, freeze rule, token refresh, tracker feature, Capacitor wrap).
What is inside	• Sections: update changelog (1-13), tracker feature follow-up (14-21), fix pass (22-30), app wrap pass (31-41).
Edit when	Add a new numbered entry for each future change.
Caution	Mentions tests/logic.test.mjs (does not exist now). Item 14 is outdated: built-in Sleep/Water trackers are no longer auto-added (BUILTIN_TRACKERS is empty).

4.2 Automation (CI)

`.github/workflows/android.yml` 2.8 KB

What it does	GitHub Actions pipeline named Build Android.
Why it exists	Runs tests automatically and builds installable Android files in the cloud, so no Android Studio is needed.
What is inside	• Triggers: push to main, any pull request, manual run (workflow_dispatch). Older runs on the same branch are cancelled. • Runs on ubuntu-24.04 with Node 22 and Java 21 (Temurin). Actions used: checkout v6, setup-node v6, setup-java v5, upload-artifact v6. • Job test: npm ci, npm test, npm audit --audit-level=high. A high severity vulnerability will turn the job red. • Job debug-apk (needs test): npm ci, npx cap sync android, gradlew assembleDebug, uploads artifact progression-debug-apk (about 3.8 MB). • Job release-aab (needs test, main branch only): runs only when GitHub secret KEYSTORE_BASE64 exists. Builds bundleRelease, signs with jarsigner, uploads artifact progression-release-aab. Without secrets the job still shows green but its steps are skipped.
Edit when	Edit to change Node/Java versions, add steps (lint, deploy), or rename artifacts.
Caution	Required secrets for signing: KEYSTORE_BASE64, KEYSTORE_PASSWORD, KEY_ALIAS, KEY_PASSWORD. Pin action versions; bump them when GitHub shows deprecation warnings.

4.3 Tests

`tests/merge.test.mjs` 1.2 KB

What it does	Tests the cloud merge and backup validator in state.js.
Why it exists	Protects against data loss when local and cloud copies are merged.
What is inside	• Checks: logs from both sides are unioned (2 keys), rest days unioned, reviews kept, achievements not duplicated, merged result passes validateBackup, newer lastModified wins.
Edit when	Add a test here whenever mergeStates or validateBackup changes.

`tests/engine.test.mjs` 4.7 KB

What it does	14 unit tests for engine.js. Added during the Oct 2026 quality pass.
Why it exists	Locks the streak, freeze, rest-day, perfect-day, comeback and rank rules so future refactors do not break them.
What is inside	• addDays across month/year; dateRange; rankForLevel thresholds; canAddRestDay window; freezeAllowance; freezeDatesForHistory (earned freezes only, skips rest days, empty input); isPerfect; comebackCount (first day, missed day, rest/freeze not a miss); timeMetrics (early bird before 8, late night 22+); average.
Edit when	Add a test before changing any rule in engine.js.
Caution	Run with npm test (needs Node.js installed). Both test files together report 15 passing.

4.4 Web app files (www/)

`www/index.html` 1.9 KB

What it does	The only HTML page. A thin shell; JavaScript builds everything else.
Why it exists	Entry point for browser and Android WebView. Holds the security policy and meta tags.
What is inside	• Meta: viewport with viewport-fit=cover (notch support), description, theme-color #0a0f1d, color-scheme dark, PWA/Apple tags, manifest and icon links, preload of IBM Plex Sans 400. • Content-Security-Policy: default-src self; script-src self (no inline scripts); style-src self + unsafe-inline; font-src self; img-src self + data; connect-src self + https (needed for Supabase); form-action none. • Body: noscript message, div#app (all UI), div#toast (role=status, notifications), dialog#dlg (modals). • Scripts (type=module): js/native.js then js/app.js.
Edit when	Edit for meta tags, title, preload, or CSP.
Caution	style-src unsafe-inline is required because the app uses inline style attributes and charts.js injects a style tag. If you add analytics or ads, you must allow their domain in the CSP. Do not turn #app into a main tag: styles.css already styles main inside #app.

`www/styles.css` 36.5 KB (177 lines, compact)

What it does	All visual design, themes, layout and animation.
Why it exists	Single stylesheet keeps the app small and cacheable.

What is inside	• @font-face for 6 self-hosted fonts. • :root design tokens: colors (--bg, --sf, --el, --ln, --tx, --ac, --ok, --wn, --er, --xp, --lv, stat colors --st0..--st6), spacing (--s1..--s6), radius, fonts (--f1 Chakra Petch for headings/numbers, --f2 IBM Plex Sans for body). • Light theme in two places: prefers-color-scheme media query and :root[data-theme=light]. Theme is chosen by app.js setting data-theme on the html element. • Safe-area padding for phone notches/gesture bar. • Layout: sidebar nav on desktop, bottom tab bar under 760 px, cards, buttons, progress bars, calendar, medals, tracker cards, level-up and achievement modals, skeleton loaders. • Later blocks (PRODUCT UI POLISH, Deep Space Blue background, Rest arc day, Light theme backgrounds) override earlier rules. • prefers-reduced-motion and data-motion=reduce switch animations off.
Edit when	Edit tokens in :root to re-theme the whole app.
Caution	Rules are repeated across blocks (for example main{} appears more than once). The LAST matching rule wins, so edit the last block when a change seems ignored.

`www/manifest.webmanifest` 0.7 KB

What it does	PWA install metadata for browsers.
Why it exists	Lets Chrome and others install the web app to the home screen with the right name, colors and icons.
What is inside	• id /progression, name and short_name Progression, lang en, start_url ./, scope ./, display standalone, orientation portrait, colors #0a0f1d, categories productivity/lifestyle/health. • Icons: icon-192.png (any), icon-512.png (any), icon-maskable-512.png (maskable).
Edit when	Edit name, colors, categories.
Caution	Android APK does not read this file; the APK name/icon come from the android/ folder.

`www/sw.js` 2.2 KB

What it does	Service worker. Makes the web version work offline.
Why it exists	Offline cache for browser users and faster loads.
What is inside	• RELEASE_CACHE = progression-v6. APP_SHELL lists every file to pre-cache (index, css, manifest, all 9 JS files, 6 fonts, 4 icons). • Install: caches the shell (one failed file does not break install), skipWaiting. Activate: deletes old caches, claims clients. • Fetch: GET only. HTML, JS, CSS and navigations are network-first with cache fallback. PNG, woff2, webmanifest are cache-first. Only successful responses are cached.
Edit when	Bump RELEASE_CACHE on every release. Add any new file to APP_SHELL.
Caution	app.js registers it only when NOT inside Capacitor (window.Capacitor undefined), because the APK already ships files locally.

`www/icon-192.png, icon-512.png, icon-maskable-512.png, apple-touch-icon.png` 12-27 KB each

What it does	App icons for the web/PWA (navy background, gold star in ring).
Why it exists	Install icons for Chrome and iOS home screen.
What is inside	• Maskable icon keeps the star inside the safe zone so Android can crop it.
Edit when	Replace with same names and pixel sizes.

`www/fonts/*.woff2 (6 files)` small

What it does	Chakra Petch 500/600/700 and IBM Plex Sans 400/500/600, Latin subset.

Why it exists	Self-hosting removes Google Fonts calls (privacy and offline) and keeps the CSP strict.
What is inside	• Both families are open licence (OFL), from @fontsource.
Edit when	Add a file and an @font-face line to add a weight.

4.5 JavaScript modules (www/js/)

`www/js/engine.js` 3.6 KB

What it does	Pure rule functions with no browser access. Shared by the app and Node tests.
Why it exists	Keeps the core scoring rules testable and independent from the UI.
What is inside	• Dates: localDate/localDateKey, parseLocalDate, addDays, dateRange (always local calendar, never UTC). • rankForLevel: S at 30, A at 20, B at 12, C at 7, D at 3, else E. • ACHIEVEMENT_CATEGORIES: Missions, Streaks, Perfect days, Hours, Levels, Progress, Goals, Time, Recovery. • canAddRestDay(days, date): false if another rest day exists within 6 days before or after. • freezeAllowance(best) = floor(best / 14). freezeDatesForHistory: walks days in order and fills single-day gaps until the allowance is spent. • isPerfect: not a rest day, at least 1 scheduled, all completed. • comebackCount: counts completions that follow a real missed scheduled day (rest and freeze days are not misses; first completion never counts). • timeMetrics: counts logs before 08:00 (early bird) and from 22:00 (late night) using each log's timestamp. • average(values).
Edit when	Edit to change a rule. Update tests/engine.test.mjs in the same commit.

`www/js/state.js` 6.5 KB

What it does	Data safety: validation, credential stripping, snapshots, merge.
Why it exists	Stops bad imports, keeps Supabase keys out of backups, and combines local and cloud data.
What is inside	• Constants: STATE_VERSION 2, BACKUP_KEY pp_bk, SNAPSHOT_KEY pp_snapshots, SUPABASE_CONFIG_KEY pp_supabase_config. • isDate: strict yyyy-mm-dd and rejects impossible dates (Feb 31). • validateBackup: checks goals, missions, logs (key must equal missionId|date), trackers, tracker logs, XP and stat rows. Backups must be v2. • get/setSupabaseConfig: URL, anon key, email, enabled flag in localStorage. • sanitizeState: deep copy with settings.supabase removed. • readSnapshots/writeSnapshot: keeps the last 7 daily snapshots. • mergeStates(local, cloud): newer lastModified wins; logs, trackerLogs, reviews, restDays, freezeDates are unioned; goals, missions, trackers, notes, achievements merged by id.
Edit when	Edit when the data shape changes. Add the new fields to validateBackup.
Caution	Deletions are not tracked, so an item deleted on one device can come back after merge. Bump STATE_VERSION only with a migration plan.

`www/js/trackers.js` 5.7 KB

What it does	Logic for trackers (sleep, water, custom).
Why it exists	Separates tracker math from the screen code.

What is inside	• TRACKER_COLORS map, BUILTIN_TRACKERS = [] (users create everything). • normalizeTrackers: cleans the list and drops unused legacy built-ins. • writeTrackerValue (0 deletes the entry; rating must be 1-5), addWater, trackerValue, dateRows. • Stats: average, goalHit, goalHitCount, currentStreak, bestStreak, sleepDebt (skips unlogged days), normalizeValue (percent of goal; lower-is-better inverted, capped 150). • makeCustomTracker (unique id, 24-char unit limit), compareSeries (up to 3 trackers, blue/yellow/red), validateTrackerDefinition. • Special cases by id: water shows glasses (250 ml each) when unit is glasses.
Edit when	Edit to add tracker types or stats.
Caution	Type and color lists are duplicated in state.js validateBackup. Change both.

`www/js/charts.js` 6.2 KB

What it does	Pure SVG trend chart builder for trackers.
Why it exists	Charts with no library: small, offline, accessible.
What is inside	• rollingAverage(rows, read, 7), renderTrendChart({series, goal, unit, rolling, summary}) returns an HTML string with SVG, gridlines, goal line, gaps for missing days, gradient area fill, glossy dots, legend and a screen-reader summary. • At load it injects a tiny style tag and global listeners so tapping or focusing a point shows a tooltip (touch devices do not show SVG title tips).
Edit when	Edit for chart look. Tooltip style lives inside this file.
Caution	Module has side effects at import time (adds document listeners once, guarded by globalThis.__trackerTip).

`www/js/supabase.js` 4.8 KB

What it does	Optional cloud sync client using plain fetch to Supabase REST and Auth.
Why it exists	Lets a user sync across devices with their own Supabase project.
What is inside	• Supabase.cfg/saveConfig, ensureSession (refreshes token 30 s before expiry; clears session only on 400/401/403), sendMagicLink (email OTP), verifyOtp (6-8 digit code), pull (select state, updated_at), push (upsert with merge-duplicates), loadSession (also reads access_token from URL hash), saveSession. • Session stored in localStorage key pp_supabase_session.
Edit when	Edit to change endpoints or auth flow.
Caution	Session token and anon key sit in plain localStorage. Planned improvement: secure storage plugin. User must type their own Supabase URL and key (planned: hosted sync).

`www/js/views.js` 0.3 KB

What it does	Tiny helpers: safeText (HTML escape) and percent (clamp 0-100).
Why it exists	Prevents HTML injection when user text is shown.
What is inside	• app.js also defines its own escape function e(); both do the same job.
Edit when	Any text from the user must pass through safeText or e() before going into a template string.

`www/js/wizard.js` 0.15 KB

What it does	Two constants for new goals: DEFAULT_GOAL_DAYS 90 and MAX_GOAL_DAYS 730.
Why it exists	Single place for goal length limits.

Edit when	Edit numbers to change limits.

`www/js/native.js` 0.9 KB

What it does	Capacitor hooks. Does nothing in a normal browser.
Why it exists	Makes the Android app feel native.
What is inside	• Status bar: dark style, #0a0f1d background, not overlaying the WebView. • Back button (needs @capacitor/app): closes the open dialog first, else goes back in history, else exits the app.
Edit when	Edit to add more native hooks (haptics, notifications) later.
Caution	Plugins are read from window.Capacitor.Plugins; wrapped in try/catch so a missing plugin never crashes the app.

4.6 Inside app.js (section map)

app.js is one large file of 259 lines (very long lines). It is organized by comment banners. Use this map to find code quickly (search for the name in the file).

Section	What it holds
UTIL	$ (querySelector), uid(prefix), e (HTML escape), clamp, num.
Dt (date engine)	iso, today, add, diff, fmt, ok, range. All local dates, format yyyy-mm-dd.
CFG	stats list (Discipline, Focus, Strength, Knowledge, Consistency, Endurance, Productivity), difficulty names/XP, bonuses, milestones, need(n) level formula.
ACH	24 achievement definitions.
blank / Store	State creation, norm (clean loaded data), validate, backup (localStorage + snapshot), export, save (reconcile streaks, stamp lastModified, debounce 700 ms push), flush (cloud push with retry 5s/15s/60s/300s), migrate (old habit tracker import from key wa_bk).
Notify	Toast messages and the notification history (last 60 in S.notes).
XP	Ledger of earned XP rows (credit, total, on, between).
Level / Stat	Level.info(totalXP) returns level, remainder, needed, percent. Stat tracks the 7 stats.
Goal	get, valid, total, elapsed, progress (planned vs done), setStatus (completed pays goal XP).
Mission	make, frac (0-1 completion), sched(date) which missions apply, set (log and settle), retire (stops from tomorrow).
Day / Score	Rest-day toggle; daily score weighted by difficulty.
Streak	reconcile (freeze allocation), calc (current/best), active, perfectDay.
Ach	metrics, check (unlock loop), revealNext, milestones (25/50/75/100%).
Engine	run(fn) wraps any change: re-evaluate milestones, achievements, save, re-render, show level-up or achievement modal. recomputeDay and settle rebuild XP/stat rows.
Stats / Arc	Chart series; the Winter Arc phases (Foundation, Build, Finish).
IC and components	Icon paths, ic, bar, hd, badge, sbadge, empty, fld, tgl, Modal (confirm, levelup, achievement).
View	Screens: home, missions, trackers, goalsPage, goal (detail), arc, progress, stats, achv, profile, settings, plus ring, chart, medal, topbar, board, ledger.
Pref / Wiz	Pref.apply sets data-theme and data-motion. Wiz is the 4-step new-goal wizard with the mission line parser: name | target | unit | stat.
ChartUX + handlers	Chart hover/pin tooltips; one big click handler keyed by data-a; change, submit, keydown handlers; swipe left/right on Today to change day.

Section	What it holds
BOOT	Load session, load pp_bk (or migrate wa_bk), render, optional cloud pull and merge with a confirm dialog when cloud is newer, register service worker (not in Capacitor).

Action names (data-a attribute) used by the click and change handlers

Group	Actions
Navigation	go, goalsgo, gv (open goal), day, dd (day +/-), open, cm (month), rng (range)
Missions	tog (check), full (complete qty), val (enter amount), rest, rm (retire), addm / addgo (add missions)
Goals	new, wn / wp / wc (wizard next/back/cancel), gs (status), gdel (delete), gf (filter)
Trackers	addTracker, editTracker, archiveTracker, deleteTracker, trackerSave, trquick, trwater, trundo, trsaveinput, trrange, trdate, compare
Modals	mno (close), myes (confirm), moment (continue after level-up)
Settings/data	theme, tgl, name, bonus, arcGoal, exj (JSON export), exc (CSV export), imp (import), snapshots / restore, reset
Sync	supabase (config fields), supabaseLogin, supabaseVerify
Charts	chartpoint (pin a point)

4.7 Android project (android/)

`android/app/build.gradle` 3.2 KB

What it does	Main Android app build file.
Why it exists	Controls app ID, version, shrinking and signing.
What is inside	• namespace and applicationId com.sanjaydigitalworks.progression, versionCode 1, versionName 1.0. • Release build: minifyEnabled true and shrinkResources true (R8 shrinks code and resources). • Signing: if android/keystore.properties exists, release is signed from it (storeFile, storePassword, keyAlias, keyPassword). Otherwise release is unsigned. • Dependencies: appcompat, coordinatorlayout, core-splashscreen, capacitor-android, junit and espresso for tests. • google-services plugin applies only if google-services.json exists (not used now).
Edit when	Raise versionCode before every Play upload.
Caution	Test the release build on a real phone after any plugin change (R8 can remove needed classes). Never commit keystore.properties or .jks files.

`android/app/src/main/AndroidManifest.xml` 1.7 KB

What it does	Declares the app to Android.
Why it exists	Permissions, launcher activity, file provider.
What is inside	• allowBackup=false (no automatic cloud backup of app data), icons, theme AppTheme. • MainActivity: launcher, singleTask, exported, handles config changes itself so rotation does not restart. • FileProvider for sharing files (paths in res/xml/file_paths.xml). • Permissions: INTERNET, POST_NOTIFICATIONS (Android 13+, for future reminders).
Edit when	Edit to add permissions or components.
Caution	Exact alarms need extra permissions if reminders must fire at exact times.

`android/app/src/main/java/com/sanjaydigitalworks/progression/MainActivity.java` 138 B

What it does	The Android activity. Extends Capacitor BridgeActivity and is empty.
Why it exists	BridgeActivity hosts the WebView that shows www/.
Edit when	Add custom native Java code here only if a plugin cannot do it.

`android/build.gradle` 0.6 KB

What it does	Top-level Gradle file.
Why it exists	Sets the Android Gradle Plugin version and repositories.
What is inside	• AGP 8.7.2, google-services classpath 4.4.2, repositories google() and mavenCentral(), clean task.
Edit when	Edit to upgrade AGP (must match the Gradle version in the wrapper).

`android/variables.gradle` 0.5 KB

What it does	Shared version numbers.
Why it exists	One place for SDK levels and library versions.
What is inside	• minSdk 23 (Android 6), compileSdk 35, targetSdk 35, AndroidX versions, junit, espresso.
Edit when	Raise targetSdk when Google Play requires it.

`android/settings.gradle, capacitor.settings.gradle` small

What it does	Lists Gradle modules.
Why it exists	capacitor.settings.gradle is GENERATED by cap sync.
What is inside	• Committed copy currently lists only capacitor-android and capacitor-status-bar. Plugins added later (app, haptics, local-notifications, preferences) are added when cap sync runs.
Edit when	Do not hand edit capacitor.settings.gradle.
Caution	CI runs cap sync before building, so CI is correct. For local Android Studio builds run npx cap sync android first.

`android/app/capacitor.build.gradle` 0.4 KB

What it does	GENERATED plugin dependency list and Java 21 setting.
Why it exists	Written by cap sync.
Edit when	Never edit by hand.

`android/app/proguard-rules.pro` 0.75 KB

What it does	R8/ProGuard keep rules.
Why it exists	Tell the shrinker what not to remove.
What is inside	• Currently only the default comments. No custom rules.
Edit when	Add -keep rules if the release build crashes on a missing class.

`android/gradle.properties`	1 KB
What it does	Gradle settings.
Why it exists	Memory and AndroidX flag.
What is inside	• org.gradle.jvmargs=-Xmx1536m, android.useAndroidX=true.
Edit when	Raise memory if the build runs out of heap.

`android/gradlew, gradlew.bat, gradle/wrapper/*`	small
What it does	Gradle launcher scripts and jar.
Why it exists	Guarantees everyone builds with the same Gradle (8.11.1).
Edit when	Change gradle-wrapper.properties distributionUrl to upgrade Gradle.

`android/app/src/main/res/ (35 files)`	images + XML
What it does	Android resources.
Why it exists	Icons, splash screens, text and themes.
What is inside	• mipmap-*/ic_launcher*.png and mipmap-anydpi-v26/*.xml: launcher icons (adaptive). • drawable-v24/ic_launcher_foreground.xml, drawable/ic_launcher_background.xml, values/ic_launcher_background.xml (#07101e). • drawable*/splash.png in portrait and landscape densities: splash screen. • layout/activity_main.xml: a WebView inside a CoordinatorLayout. • values/strings.xml: app_name, title_activity_main, package_name, custom_url_scheme. • values/styles.xml: AppTheme, AppTheme.NoActionBar, AppTheme.NoActionBarLaunch (uses the splash). • xml/file_paths.xml: FileProvider paths.
Edit when	Replace images keeping names. Edit strings.xml to rename the app.

`android/app/src/test and androidTest (2 Java files)`	tiny
What it does	Template unit and instrumented test stubs (package com.getcapacitor.myapp).
Why it exists	Left over from the Capacitor template. Not used.
Edit when	Safe to delete or replace with real tests.

5. Data and game rules

Storage keys (browser localStorage)

Key	Holds
`pp_bk`	Main app state (goals, missions, logs, XP, stats, achievements, trackers, settings). JSON, version 2. Credentials never included.
`pp_snapshots`	Last 7 daily copies of the state, keyed by date.
`pp_supabase_config`	Supabase URL, anon key, email, enabled flag. Local only, never exported.
`pp_supabase_session`	Supabase access and refresh tokens.
`wa_bk`	Old habit tracker data. Read once for migration into a goal named Imported goal.

State object (saved in pp_bk)

Field	Content
`v`	2 = STATE_VERSION
`user`	{name}
`goals[]`	id, name, desc, cat, start, end, why, success, target, current, unit, priority, status (draft/active/paused/completed/archived), ms[] (milestones hit), stoppedOn
`missions[]`	id, goalId, name, type (check / qty / dur), target, unit, diff (1-3), xp, stats[], start, end, active
`logs{}`	key missionId|date -> {id, missionId, goalId, date, value, at (timestamp ms)}
`xp[]`	ledger rows {id, key, type, src, amount, date, desc}. Types: MISSION, DAILY_BONUS, MILESTONE, GOAL_COMPLETED, ACHIEVEMENT
`stats[]`	{key, date, stat, amt}
`ach[]`	{id, date} unlocked achievements
`reviews{}`	daily review by date
`notes[]`	notification history, max 60
`trackers[]`	id, name, unit, type (number/count/duration/rating), dailyGoal, direction (higher/lower), color, archived
`trackerLogs{}`	key trackerId|date -> {trackerId, date, value}
`settings`	restDays{}, freezeDates{}, freezeBank, theme, motion, quiet, arcGoalId, bonus{day,ms,goal}, sleepGoal, waterGoal, waterUnit
`lastModified`	ISO time; decides which copy wins in cloud merge

Game rules

Rule	How it works
Mission types	check = done or not. qty = number toward a target. dur = minutes toward a target (counts toward hours achievements).
Mission XP	Default by difficulty: Easy 40, Standard 75, Hard 120. qty and dur missions pay in proportion (partial credit).
Daily score	Difficulty-weighted share of targets reached for scheduled missions. Rest days and empty days have no score.
Perfect day	Every scheduled mission complete, not a rest day. Pays the daily bonus (default 100 XP).
Milestones	At 25, 50, 75, 100 percent of a goal: 500 XP each (default).

Rule	How it works
Goal completion	2000 XP (default), applies when the goal is marked Complete.
Level	XP needed for level n = round(1000 x 1.18^(n-1) / 50) x 50. Level 1 needs 1000, level 2 needs 1200 more, and so on.
Rank	E (levels 1-2), D (3-6), C (7-11), B (12-19), A (20-29), S (30+).
Streak	A day counts when at least one mission is fully done, OR it is a rest day, OR it is freeze-protected. Current streak counts back from today (or yesterday if today is not done yet).
Rest day	Protects the streak. One rest day per 7-day span: another rest day cannot fall within 6 days before or after.
Streak freeze	One freeze earned per 14 days of best streak. Freezes fill single-day gaps in order, including yesterday.
Comeback	Finish a mission after a real missed scheduled day (rest and freeze days do not count).
Early bird / late night	Uses the time stored with each completed log. Before 08:00 or from 22:00. Old logs without a timestamp do not count.
Achievement tiers	By XP reward: Common under 200, Rare 200+, Epic 500+, Legendary 1000+.
Stats	7 stats. Missions add their difficulty value to their linked stats when fully done. Perfect day adds 1 Consistency. Active day adds 1 Discipline.
Winter Arc	A view of the selected goal in three phases: Foundation, Build, Finish (each a third of the goal length).

XP needed per level (sample)

Level	XP to finish this level	Total XP before it
1	1,000	0
2	1,200	1,000
3	1,400	2,200
5	1,950	5,250
7	2,700	9,500
10	4,450	19,150
12	6,200	28,850
15	10,150	50,950
20	23,200	123,500
25	53,100	289,550
30	121,500	669,450

All 24 achievements

ID	Name	Requirement	XP	Category
`first`	**First step**	Complete your first mission	50	Missions
`c7`	**Consistent**	7-day streak	150	Streaks
`c14`	**Two weeks strong**	14-day streak	300	Streaks
`c30`	**Unstoppable**	30-day streak	600	Streaks
`c60`	**Iron will**	60-day streak	1200	Streaks
`p5`	**Flawless five**	5 perfect days	200	Perfect days

ID	Name	Requirement	XP	Category
p20	**Perfect form**	20 perfect days	500	Perfect days
p50	**No wasted days**	50 perfect days	1000	Perfect days
f50	**Focus mode**	50 duration hours	500	Hours
f100	**Deep work**	100 duration hours	1000	Hours
l5	**Level 5**	Reach level 5	150	Levels
l10	**Level 10**	Reach level 10	300	Levels
l20	**Level 20**	Reach level 20	750	Levels
xp1	**Centurion**	Earn 100,000 XP	1000	Progress
xp5	**Quarter million**	Earn 250,000 XP	2000	Progress
fin	**Finisher**	Complete 1 goal	500	Goals
g3	**Three chapters**	Complete 3 goals	1000	Goals
g5	**Seasoned builder**	Complete 5 goals	1500	Goals
early	**Early bird**	1 early mission	150	Time
early7	**Morning protocol**	7 early missions	400	Time
late	**Late-night operator**	1 late mission	150	Time
late7	**Night shift**	7 late missions	400	Time
comeback	**Comeback**	1 comeback	250	Recovery
comeback3	**Return to form**	3 comebacks	700	Recovery

Mission line format (new goal wizard and Add missions)

```
name | target | unit | stat

Study JavaScript | 60 | min | Knowledge     (duration mission, Standard)
Read pages | 20 | pages | Knowledge         (quantity mission)
Cold shower!                                   (checkbox, ! = Hard)

Leave target empty = checkbox. Unit min/hours = duration mission.
```

6. Procedures (Windows)

A. Run the web app on your PC

1. Install Python (or Node). In cmd: cd /d "F:\!DigitalProduct\DailyActivityWebsite_mobile app\progression"
2. python -m http.server 4173 --directory www
3. Open http://localhost:4173/ . After editing, press Ctrl+Shift+R (hard refresh) because the service worker caches files.

B. Run tests

1. Install Node.js LTS from nodejs.org, then reopen cmd (npm was not installed on this PC at the time of writing).
2. npm install (first time)
3. npm test (expect 15 passing)

C. Save changes to GitHub

1. git status
2. git add .
3. git commit -m "short message"
4. git push
5. Then open GitHub, Actions tab, and wait for green checks. Red means open the run and read the failing step.

D. Get an installable APK

1. Push to main. Open Actions, click the latest run, scroll to Artifacts, download progression-debug-apk, unzip, copy app-debug.apk to the phone, allow unknown apps, install.

E. Make a signed release for Google Play

1. Create keystore once: keytool -genkeypair -v -keystore progression-release.jks -alias progression -keyalg RSA -keysize 2048 -validity 10000
2. BACK UP the .jks file and passwords in two safe places. Losing it means you cannot update the app. Never commit it.
3. Convert to text (PowerShell): [Convert]::ToBase64String([IO.File]::ReadAllBytes("progression-release.jks"))
4. GitHub repo, Settings, Secrets and variables, Actions: add KEYSTORE_BASE64 (the text), KEYSTORE_PASSWORD, KEY_ALIAS (progression), KEY_PASSWORD.
5. Raise versionCode in android/app/build.gradle, commit, push to main.
6. Download artifact progression-release-aab from the run and upload the .aab in Play Console.
7. Recommended: enable Play App Signing in Play Console.

F. Release a web/PWA update

1. Change RELEASE_CACHE in www/sw.js (progression-v6 to v7).
2. Add a numbered entry to CHANGELOG.md.
3. Push. Browser users get the new version on next load.

G. Add a Capacitor plugin

1. npm install @capacitor/plugin-name
2. npx cap sync android
3. Use it from JavaScript through window.Capacitor.Plugins (see native.js).

4. Commit package.json, package-lock.json and the updated android/ files.
5. Test a release build on a phone.

7. Known issues and technical debt

Honest list as of commit ac7343d. Fix High items first.

Level	Issue	Detail / plan
High	**Keystore files are not git-ignored**	Add *.jks, *.keystore, keystore.properties to .gitignore before creating a release key. If a key was ever committed, treat it as leaked.
High	**Data only in localStorage**	Browser storage can be cleared by the OS or user. Plan: Capacitor Preferences or SQLite, plus one-tap backup/restore, plus a tested migration from pp_bk.
High	**Supabase token and anon key in plain localStorage; user must type URL and key**	Plan: secure storage and a hosted sync, or hide the sync section until ready.
Medium	**app.js is one 84 KB file**	Hard to read and test. Split into modules (store, xp, goals, streaks, views, actions). Do it with the engine tests as safety net.
Medium	**Plugins installed but unused**	@capacitor/haptics, local-notifications, preferences are in package.json but no code calls them. Reminders feature not built yet.
Medium	**Two signing paths**	build.gradle can sign via keystore.properties; the CI signs the AAB with jarsigner afterwards. Keep one.
Medium	**Cloud merge cannot delete**	Deleted goals or missions can return after merge. Needs tombstones (deleted ids).
Medium	**Little test coverage**	Only engine.js and mergeStates/validateBackup are tested. trackers.js, charts.js, supabase.js, app.js are not.
Medium	**R8 shrinking not yet tested on a device**	minifyEnabled is true. Install a release build and test reminders, status bar, back button, import/export.
Low	**Stale docs**	README test command, CHANGELOG mentions tests/logic.test.mjs, CHANGELOG item 14.
Low	**Generated Gradle files out of date locally**	capacitor.settings.gradle lists only status-bar until cap sync runs.
Low	**versionCode fixed at 1**	Must be raised for every Play upload.
Low	**No accessibility audit**	Check contrast 4.5:1, tap targets 48 dp, TalkBack labels, 200 percent text.
Low	**sw.js file list is manual**	A new JS file not added to APP_SHELL will not be cached for offline browser use.
Low	**Repository visibility**	Check Settings, Danger Zone: the repo was still public when last checked.

8. Roadmap status and glossary

Roadmap (from the Notion plan: Progression App, Polish, Launch, Sell)

Phase	Scope	Status
Phase 0 Decide	Pick a niche, final name and app ID, brand colors.	Not done
Phase 1 Foundation	allowBackup off, R8 on, release signing wired, CI, tests. Still open: durable storage, secure tokens, remove manual Supabase setup.	Partly done
Phase 2 Features	Onboarding and templates, reminders, haptics and animations, widget, backup/restore.	Not done (plugins installed)
Phase 3 Quality	Engine tests added, CI audit added. Still open: split app.js, accessibility audit, crash reporting.	Partly done
Phase 4 Monetization	Free vs Pro, paywall, Play Billing or RevenueCat.	Not started
Phase 5 Launch	Play Console, closed testing, store listing, privacy policy, Data Safety form.	Not started
Phase 6 Grow and sell	Marketing, metrics for 2-3 months, then sell with revenue proof.	Not started

Suggested order from here: split app.js with tests as safety net, durable storage plus secure token, release signing test on a phone, reminders and onboarding, then Pro and billing.

Glossary

Term	Meaning
PWA	Progressive Web App: a website that can be installed and work offline.
Capacitor	Tool that wraps a web app inside a native Android (or iOS) app.
WebView	The Android component that shows web pages inside an app.
APK / AAB	APK is an installable Android file. AAB is the upload format for Google Play.
R8	Android tool that shrinks and obfuscates code in release builds.
Keystore	File holding your private signing key. Required to publish and update on Play.
CI	Continuous integration: GitHub Actions running tests and builds on every push.
Service worker	Browser script that caches files for offline use.
CSP	Content Security Policy: browser rules limiting which scripts and hosts a page may use.
OTP	One-time password code sent by email for Supabase login.
RLS	Row level security in Supabase: each user can only read and write their own row.
XP ledger	The list of XP rows (S.xp); total XP is the sum.
Settle	Recalculate XP and stat rows for one mission on one day.

End of document. Update this reference when files are added, renamed or rules change.